

AL-KHIDMAT FOUNDATION · ALIBABA · GITHUB

Mustahiq AI

Finding, verifying and fairly allocating aid — **and helping beneficiaries grow.**

A staff decision-support layer for Al-Khidmat's beneficiary programmes, plus a no-fee marketplace for micro-entrepreneurs.

Team submission · built on free infrastructure · everything shown is real and running

Deciding who gets limited help is manual — and hard to defend



A household often qualifies for several programmes

but a field officer only has time to check one or two, so real needs are missed.



“Why them and not me?” has no clear answer

prioritisation lives in spreadsheets and memory; two officers rank the same people differently.



Verification, waiting lists and budgets are tracked by hand

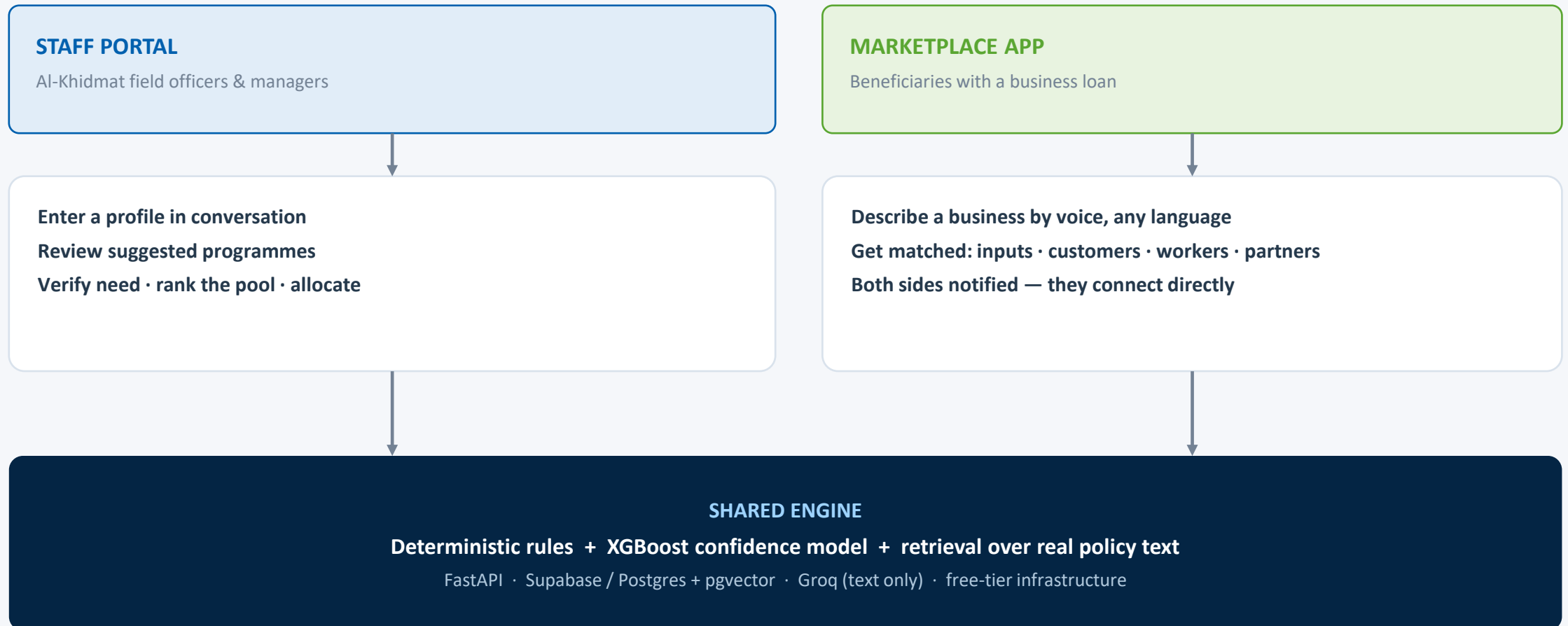
across programmes and cycles — slow, and easy to lose.



A beneficiary who starts a business is on their own

the loan ends; nobody connects them to suppliers, customers or partners.

One platform, two sides, a shared engine



Eligibility: a five-stage pipeline, humans in control



Rules ELIMINATE. The model only RANKS confidence that verification will confirm need.

No LLM anywhere in scoring · no automatic enrolment · a failed verification simply exits

Every ranking is explainable — to the factor

THE RUBRIC, SHOWN TO STAFF · one example

Income (inverse)	30%
Dependents	20%
Disability	15%
No prior assistance	20%
School-age children	15%

Weights are set per programme and snapshotted with every cycle.

STRUCTURAL FAIRNESS

How someone entered — staff-identified or walk-in — is recorded for audit and can never enter a ranking calculation.

PLAIN-LANGUAGE REASONS

Every suggestion and every rank comes with a sentence a caseworker can read aloud — traceable to the real policy text it came from.

Calibrated, auditable, deliberately narrow

WHAT THE MODEL IS

XGBoost — gradient-boosted trees

Trained on 15,000 synthetic profiles

~54,000 labelled (profile, programme) rows

Held-out ROC-AUC ~0.76 · F1 ~0.71

Predicts one thing: will verification confirm need?

WHAT IT DOES NOT DO

No LLM in the scoring path — deterministic

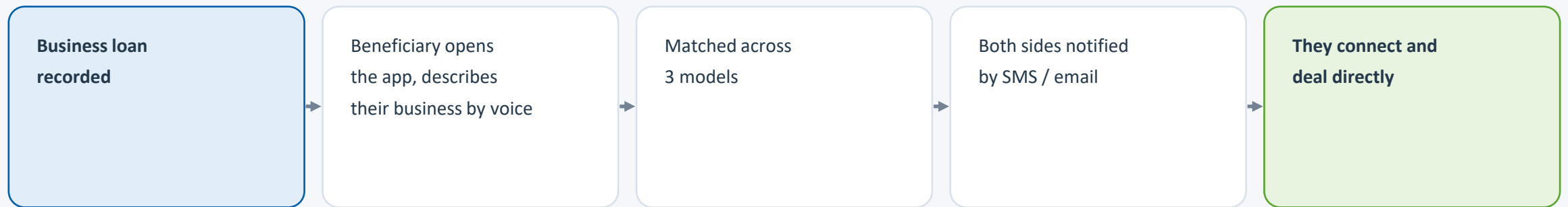
No training on real beneficiary data

Never enrolls anyone — staff decide

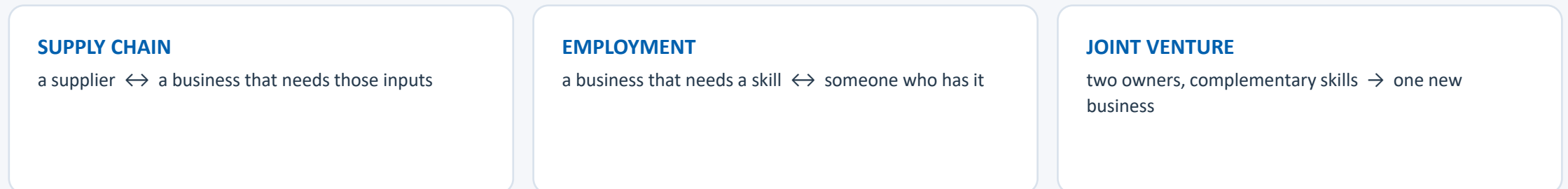
Never a funding priority — that's the rubric

Confidence shown only as low / medium / high

The marketplace: from a loan to a network



MATCHED ACROSS 3 MODELS



No fees, ever. No commission. Al-Khidmat introduces only — terms and pricing are between the two businesses.

It's built — not mocked

Both web apps, the API, the engine and the full workflow run today.

STAFF PORTAL

Register → discovery worklist → outreach → verification → ranked candidate pool → allocation → department reports

add screenshot: staff dashboard

MARKETPLACE APP

Phone + SMS-code login → voice / text listing → review → match results → in-app chat to arrange details

add screenshot: match results

Tested end to end — 41 eligibility tests, 33 staff-workflow tests, marketplace smoke tests, all green.

Built for the real constraints

Free tier only

No GPU. Local CPU embeddings, hosted rules + trees, Groq free tier for text only.

Every answer is cited

Retrieval runs over uploaded policy documents — never the open web.

Any language in, English pipeline

A beneficiary speaks Urdu or Punjabi; matching still works.

Staff-operated by design

Beneficiaries never log into the eligibility side. The marketplace is the deliberate exception.

Two scheduled jobs

Bi-weekly ranking cycle and a daily marketplace cleanup — the only things not real-time.

Rate-limit aware

One Groq wrapper, live token logging, automatic fallback key on a 429.

WHAT CHANGES FOR AL-KHIDMAT

Faster, consistent, defensible aid — and a path to self-reliance

BEFORE

One programme checked per visit

Prioritisation by memory & spreadsheet

“Why them?” — no record

Loan ends, beneficiary alone

WITH MUSTAHIQ AI

Every active programme, in milliseconds

One rubric, snapshotted, factor-by-factor

A plain-language reason for every decision

Matched to a network that keeps their business going

Roadmap: real verification data → model retraining loop · row-level security · production auth & payments

From aid, to auditable aid, to self-reliance.

Mustahiq AI · AI-Khidmat × Alibaba × GitHub hackathon

Repository: github.com/zobiahussain/mustahiq-ai

Thank you.